

Problem Statement ID : SIH26018

Problem Statement Title : Intelligent Land Record Digitization and Validation System

Theme : Smart Automation

PS Category : Software

Team ID : SIH26-A0H-T309

Team Name (Registered on portal) : SIXTH SENSE



One evidence-driven layer for citizens, government officers and future property discovery

EXTRACT → VALIDATE → RECONCILE → EXPLAIN → PRIORITISE → ACT

PROPOSED SOLUTION

Problem • prototype • differentiation • key capabilities

PROBLEM AT HAND

- FRAGMENTED EVIDENCE**
Deeds, RoR, mutation, registration, maps and legacy files can exist in different records/systems.
- HUMAN INTERPRETATION GAP**
A citizen may possess documents but still cannot tell whether they agree or what a mismatch means.
- MANUAL OFFICER WORK**
Staff must compare fields, dates and spatial evidence repeatedly before deciding which cases deserve attention.
- MARKET INFORMATION ASYMMETRY**
Buyers depend on brokers, local claims or advertisements because property evidence is difficult to understand and compare.

OUR SOLUTION — WEB PROTOTYPE / DEMO

PROPERTY INTELLIGENCE / P-00421
92% READY

2.50 acres • Survey 142/3 • Coimbatore

	RoR	SALE DEED	EC	MAP
FIELD	SOURCE VALUE			
Owner		Ravi Kumar	MATCH	Identity aligned
Survey		142/3	MATCH	Parcel ID aligned
Area		2.50 vs 2.31 ac	REVIEW	7.6% difference
Timeline		2019→2023	MATCH	No break detected

AI EXPLANATION
 “Survey area is 7.6% lower. This may indicate a measurement, subdivision/update or boundary-record issue. Verify the latest survey evidence.”

WHY WE STAND OUT / KEY FEATURES

- 01 Property Intelligence Profile**
One structured profile links owner, parcel, documents, events and spatial evidence.
- 02 Evidence-linked AI**
Every important finding points back to the fields/documents that produced it.
- 03 Conflict reasoning**
Not just “mismatch”: what differs, where, how much, likely cause, severity and next check.
- 04 Citizen + Officer modes**
Same evidence engine; simple explanation for citizens, investigation queue for officers.
- 05 Marketplace-ready trust**
Verified attributes become the basis for buyer–property matching instead of only price/ads.

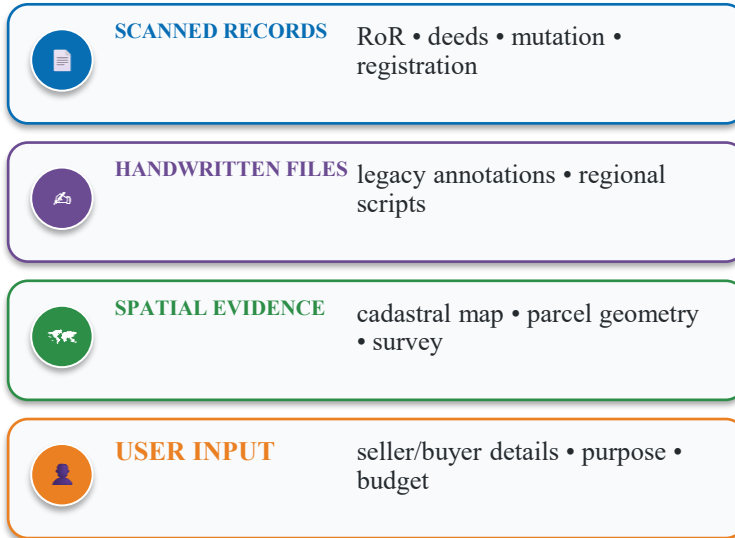
MARKETPLACE — SAME AI ENGINE, NEW DECISION

- 1 SELLER**
Create property profile
- 2 EVIDENCE**
Upload/link records
- 3 VERIFY**
AI checks consistency
- 4 LIST**
Publish verified attributes
- 5 BUYER**
Describe needs in natural language
- 6 MATCH**
AI ranks fit + evidence
- 7 UNDERSTAND**
Buyer sees risks & reasons
- 8 CONNECT**
Direct buyer ↔ seller

TECHNICAL APPROACH

Complete technical architecture and technology stack

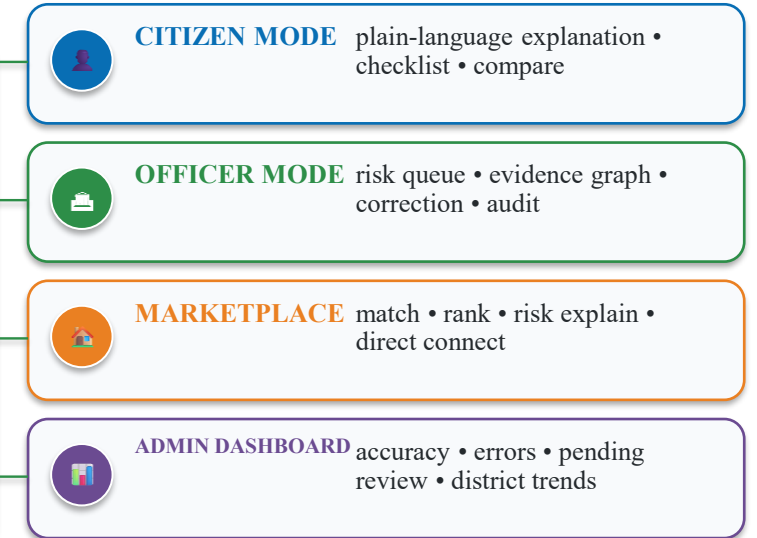
DATA / EVIDENCE INPUTS



PROPERTY TRUTH ENGINE



DECISION / USER OUTPUTS



FRONTEND

React / Next.js • responsive web UI • maps • dashboards • role-based screens

BACKEND

Python / FastAPI • validation APIs • inference orchestration • audit services

DATA

PostgreSQL • PostGIS • object storage • graph model • versioned evidence

AI / SECURITY

OCR + vision/LLM • confidence • RBAC • encryption • audit trail • human review

FEASIBILITY AND VIABILITY

Technical feasibility • implementation • safety • before / after

TECHNICAL FEASIBILITY

- ✓ OCR / Computer Vision / NLP / GIS are commercially available building blocks.
- ✓ Deterministic rules handle measurable checks; AI handles classification and explanation.
- ✓ Small labelled prototype can demonstrate the complete pipeline.
- ✓ Spatial comparison can begin with sample parcel polygons.

DATA ACCESS FEASIBILITY

PROTOTYPE → permitted/public/synthetic records for the demo.
 PILOT → authorised departmental datasets / APIs.
 PRODUCTION → connect to existing government LRMS, DILRMP, GIS, registration and other authorised databases through approved APIs/data-sharing.
 The platform is an intelligence layer on top of official records not a replacement database.

ECONOMIC / OPERATIONAL VIABILITY

FACTOR	PROTOTYPE	SCALE-UP
Compute	Local / cloud	GPU / cloud
Data	Sample corpus	Authorised feeds
Review	Human queue	District workflow
Storage	Object store	Gov-grade infra
Revenue	Pilot / grant	SaaS / API / services

VALUE = fewer manual comparisons + faster exception review + clearer citizen decisions

IMPLEMENTATION VIABILITY

P1 sample corpus → P2 extraction + validation → P3 citizen/officer workflows → P4 marketplace → P5 authorised integration and district pilot.

MAINTENANCE / SAFETY

- Confidence threshold + human escalation.
- Source-linked evidence prevents unsupported claims.
- Versioning preserves historical records.
- RBAC separates citizen/officer/admin access.
- Audit trail logs AI output + human correction.
- AI result is decision support, not legal title certification.

BEFORE → AFTER

Open multiple records	—	One evidence profile
Read manually	—	AI extracts
Compare fields	—	Engine reconciles
Find mismatch	—	AI explains conflict
Decide what to do	—	Officer verifies exception

IMPACT AND MARKET VALUE

Impact • India scale • government value • marketplace

INDIA SCALE

6.30 LAKH

villages / CLR

4.02 CRORE

digital maps

4,148

SROs computerised

DILRMP 3.0

2026–2031

Official DoLR dashboard / programme

COMMON PEOPLE

- Understand property evidence in plain language.
- See exactly what disagrees before committing money.
- Reduce dependence on broker/local claims.
- Compare properties by trust + suitability, not only price.

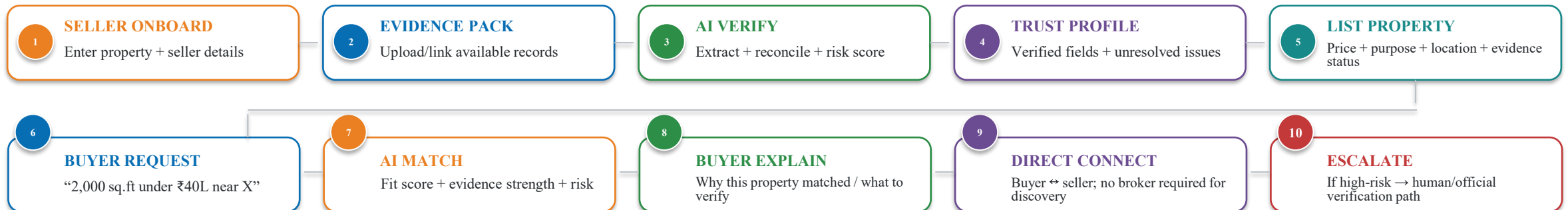
GOVERNMENT / OFFICERS

- Exception-first review queue.
- Evidence-linked case reasoning.
- Less repetitive comparison and data entry.
- Analytics on recurring discrepancy patterns and record quality.

ECOSYSTEM / BUSINESS

- Banks: structured due diligence.
- Legal: property timeline/evidence organisation.
- Developers: acquisition screening.
- Government: intelligence/API layer for modernization.

MARKETPLACE — COMPLETE END-TO-END PROCESS



SCALE / BUSINESS MODEL

District pilot → State platform → National integrations | Gov/API licensing • institutional subscriptions • verification services • marketplace discovery

INTELLIGENT DIGITALIZATION

Deep dive: capture → preprocess → OCR/HTR → field intelligence → confidence → human review

01 CAPTURE

PDF / scan /
photo

02 PREPROCESS

deskew • denoise
crop • contrast

03 LAYOUT AI

detect tables,
regions, fields

04 OCR + HTR

printed + handwriting
multiple Indian languages

05 FIELD AI

owner • survey • area
mutation • registration

06 CONFIDENCE

field-level score
uncertain → review

INPUT PROBLEM — REAL RECORDS ARE MESSY

Example source:

- 1980s handwritten register
- faded ink + stamps + folds
- Tamil / regional terminology
- inconsistent column layouts
- abbreviations and overwritten values
- same person may appear with spelling variations

A normal scanner only creates another image. Our system creates usable evidence.

WHAT OUR AI ACTUALLY DOES

1. Image quality assessment → repair only when useful.
2. Layout model identifies title, table, row and field regions.
3. OCR/handwriting model reads text in the detected regions.
4. NLP maps raw text to a standard land schema.
5. Entity resolution links spelling variants to candidate identities.
6. Unit/date/ID normalization creates comparable values.
7. Each field receives a confidence score + source location.
8. Low-confidence values enter human verification.

BETTER THAN ORDINARY DIGITIZATION

ORDINARY OCR

Image → text → database

Risk: errors become invisible structured data.

OUR PIPELINE

Document → structured field → confidence → evidence link → validation → explanation

Key difference:

We preserve the connection between a digital value and the exact source evidence that produced it.

Prototype output: side-by-side source document + extracted form + highlighted uncertain fields.

EXAMPLE “2.50 ACRES” → FIELD VALUE 2.50 → UNIT ACRES → SOURCE BOX [page 3, row 18] → CONFIDENCE 96% → READY FOR VALIDATION

VERIFICATION AND VALIDATION

Deep dive: rules → cross-record → temporal → spatial → duplicate reasoning → action

PROPERTY P-00421=>

OWNER

Ravi Kumar

SURVEY

142/3

AREA

2.50 acres

STATUS

REVIEW

BUSINESS RULES

Area > 0 • valid survey format • dates logical • mandatory fields

CROSS-RECORD

Owner / survey / area / classification / mutation agreement

TEMPORAL

Sale → registration → mutation → current record chronology

SPATIAL

Recorded area vs mapped area • overlap • parcel geometry

DUPLICATE / ENTITY

Duplicate survey IDs • repeated owners • possible duplicate records

FROM MISMATCH TO REASON

OBSERVED
Record = 2.50 acres
Map = 2.31 acres
Difference = 0.19 acres
≈ 7.6% lower

POSSIBLE EXPLANATIONS

- outdated geometry
- subdivision/update
- boundary measurement difference
- extraction error

AI must not choose a legal conclusion without evidence.

ACTIONABLE OUTPUT

SEVERITY
REVIEW — material but explainable candidate

EVIDENCE

- record page + extracted field
- map geometry
- relevant transaction/mutation event

NEXT CHECK
“Verify the latest survey / subdivision evidence before relying on the area.”

OFFICER QUEUE
High-impact + low-confidence cases rise to the top.

AUDIT
AI finding + source + human decision are stored together.

VALIDATION ENGINE = CHECKS RULES + CROSS-SOURCE COMPARISON + SPATIAL/TEMPORAL REASONING + GIVES CONFIDENCE RATING

AI SUPPORT ≠ LEGAL CERTIFICATION

MARKETPLACE

Deep dive: seller → evidence → verification → trust profile → AI matching → direct discovery

SELLER / PROPERTY SIDE

1 ONBOARD Seller enters property basics
location • size • price • purpose

2 EVIDENCE PACK Upload available documents
and spatial evidence

3 AI VERIFY Extract → reconcile → confidence
→ unresolved issues

4 TRUST PROFILE Verified attributes + risk flags
+ evidence strength

5 LIST Property becomes searchable
with transparent evidence status

MATCH

BUYER / REQUIREMENT SIDE

A DESCRIBE NEED “2,000 sq.ft near X
under ₹40 lakh for home”

B NORMALISE AI converts natural language
into structured preferences

C MATCH Location + size + budget + purpose
+ evidence quality

D RANK Fit score + risk + evidence strength
—not just advertisement ranking

E EXPLAIN “Why this matched” + what buyer
still needs to verify

AFTER MATCH

DIRECT CONNECT → buyer and seller communicate → buyer sees evidence/risk explanation → high-risk cases can be escalated for official/human verification → transaction-stage checks preserve the evidence trail

UNIQUE VALUE: VERIFIED ATTRIBUTES BECOME THE TRUST SIGNAL OF THE MARKETPLACE — THE SAME DATA SERVES GOVERNMENT WORKFLOW AND CITIZEN DISCOVERY.

INDIA LAND-RECORD ECOSYSTEM

6.30 LAKH
villages / CLR

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RESEARCH GAP

Digitisation alone stores records. Sixth Sense adds an evidence-linked intelligence layer that reconciles conflicts, explains them and routes action.

REFERENCES

01	SIH 2026 PS SIH26018	Intelligent Land Record Digitization and Validation System
02	Department of Land Resources	DILRMP — objectives, components, implementation
03	DILRMP 3.0 Operational Guidelines	2026–2031 programme phase
04	DoLR Current Dashboard	CLR / digital maps / SRO computerisation indicators
05	Bhoomi Samman	RoR, cadastral maps, registration and integration components

TECHNICAL BASIS / EVALUATION PLAN

OCR / DOCUMENT AI	Printed + handwritten recognition; layout-aware field extraction.
COMPUTER VISION	Map/image interpretation, seals/layout features and visual evidence.
NLP / LLM	Field classification + evidence-grounded natural-language explanation.
POSTGIS / GIS	Parcel geometry, overlap, boundary and area comparison.
KNOWLEDGE GRAPH	Property ↔ person ↔ document ↔ transaction/event relationships.
HUMAN-IN-LOOP	Confidence thresholds route uncertain/high-impact cases for review.
EVALUATION	Field accuracy • conflict precision/recall • calibration • explanation correctness • review-time reduction

LINKS

[SIH26018](#)

[DoLR / DILRMP](#)

[DILRMP 3.0 Guidelines](#)

[SIH26018](#)

[DoLR / DILRMP](#)

[DILRMP 3.0](#)